

3 Problem formulation

3.1 Problem setting and assumptions

We consider a multi-story warehouse in which autonomous mobile robots (AMRs) transport orders between source and destination floors using shared elevators. An AMR travels to an order's source, collects the load, and delivers it to the destination. The same AMR serves the order through delivery and rides an elevator whenever a movement crosses floors. The orders released together form a wave. Their origins, destinations, and ready times determine the vertical transport requests generated during wave execution.

The warehouse management system selects a wave of fixed size from a finite pool of candidate waves. Resource quantities and operational rules are given, so the decision concerns which orders enter the wave. The main formulation selects a structural class, defined by the floor distribution and directional balance of its candidate waves, and implements that choice through a fixed within-class selection rule. Direct selection of a concrete candidate wave provides a finite-pool benchmark.

Each decision instance concerns one isolated wave released at time t_w . Every available order has a known source floor, destination floor, ready-time offset, and immutable identifier. An order with ready-time offset $r_o \geq 0$ becomes eligible for processing at $t_w + r_o$. All AMRs and elevator resources are initially available at a common floor f^0 , with no outstanding reservations. The evaluation ends when all selected orders have been delivered; no other wave shares its resources during this interval.

Intra-floor movement and handling are represented by fixed pickup and drop-off service durations. Elevator travel time is proportional to the number of floors traversed, and each trip includes loading and unloading. The elevator evaluators differ in their representation of capacity and trip-time variability. They map each candidate wave to its completion time under the same wave-level decision conditions.

3.2 Notation

Tables 3.1 and 3.2 summarize the basic indices and input parameters. Candidate-wave descriptors and operational state quantities are introduced with their definitions in Sections 3.3 and 3.4.

Table 3.1. Basic indices and sets

Symbol	Definition
$f \in \mathcal{F} = \{1, \dots, F\}$	warehouse-floor index
$o \in \mathcal{O}$	order index in the available order pool
$a \in \mathcal{A} = \{1, \dots, A\}$	AMR index
$e \in \mathcal{E} = \{1, \dots, E\}$	physical-elevator index
$v \in \mathcal{V}_i = \{1, \dots, Ec\}$	virtual single-request elevator-slot index under M_i
$k \in \mathcal{K} = \{1, \dots, K\}$	candidate-wave index, shared by all evaluators
$q \in \mathcal{Q}$	selectable structural-class label
$m \in \mathcal{M} = \{M_1, M_2, M_3\}$	elevator-evaluator index

Table 3.2. Input parameters

Symbol	Definition
F	number of warehouse floors
A	number of AMRs
E	number of physical elevator cars

c	maximum number of AMRs sharing one physical elevator trip
n	fixed number of orders in a wave
K	number of candidate waves in the finite pool
t_w	decision and release epoch of the wave
$f^0 \in \mathcal{F}$	common initial floor of the AMRs and elevator resources
t_o	immutable and unique identifier of order o
$s_o, d_o \in \mathcal{F}$	source and destination floors of order o
$r_o \geq 0$	order-ready offset relative to t_w
$\gamma > 0$	elevator travel time per traversed floor
$\tau^L, \tau^U > 0$	elevator loading and uploading durations
$\tau^P, \tau^D > 0$	AMR pickup and drop-off service durations
$\alpha \in (0, 0.5)$	tail probability for structural-class construction
$\sigma > 0$	lognormal phase-time noise parameter under M_3

The resource counts, capacity, and candidate-pool size are positive integers, and $1 \leq n \leq |\mathcal{O}|$. All service durations and time offsets use the same time unit.

3.3 Candidate waves and structural classes

3.3.1 Candidate waves

Each candidate $k \in \mathcal{K}$ is an unordered set $W^k \subseteq \mathcal{O}$ containing n orders. The finite pool is generated before selection and held fixed across evaluators. Its order membership is represented by the incidence parameter δ_{ok} :

$$|W^k| = n, \quad \delta_{ok} = \begin{cases} 1, & o \in W^k, \\ 0, & o \notin W^k, \end{cases} \quad o \in \mathcal{O}, k \in \mathcal{K} \quad (1)$$

3.3.2 Wave descriptors

The floor distribution of a wave is measured using both endpoints of every order. For a candidate wave W , let $p_{f(W)}$ be the proportion of its $2|W|$ endpoints located on floor f :

$$p_f(W) = \frac{1}{2|W|} \sum_{o \in W} [\mathbb{I}(s_o = f) + \mathbb{I}(d_o = f)], \quad f \in \mathcal{F} \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function. Endpoint dispersion is represented by the entropy

$$C(W) = - \sum_{f \in \mathcal{F}} p_f(W) \log p_f(W), \quad 0 \log 0 = 0 \quad (3)$$

Here $\log$ denotes the natural logarithm. Larger C reflects greater diversity in the endpoint distribution across floor labels, arising from broader support or a more even distribution. It describes how endpoints are distributed among floors; travel distances are determined by the numerical differences between the relevant floor indices. Directional imbalance measures the asymmetry between upward and downward order movements. Their counts are

$$N^\uparrow(W) = \sum_{o \in W} \mathbb{I}(d_o > s_o), \quad N^\downarrow(W) = \sum_{o \in W} \mathbb{I}(d_o < s_o). \quad (4)$$

The absolute difference is normalized by the number of cross-floor orders:

$$I(W) = \begin{cases} \frac{|N^\uparrow(W) - N^\downarrow(W)|}{N^\uparrow(W) + N^\downarrow(W)}, & N^\uparrow(W) + N^\downarrow(W) > 0, \\ 0, & N^\uparrow(W) + N^\downarrow(W) = 0. \end{cases} \quad (5)$$

Thus $I \in [0,1]$, with zero indicating balanced cross-floor counts or the absence of cross-floor orders, and one indicating movement in only one direction. Same-floor orders do not contribute to either count. Ready-time dispersion measures how order eligibility is spread within the wave. With $\bar{r}_W = |W|^{-1} \sum_{o \in W} r_o$, define

$$T(W) = \begin{cases} \frac{\sqrt{|W|^{-1} \sum_{o \in W} (r_o - \bar{r}_W)^2}}{\bar{r}_W}, & \bar{r}_W > 0, \\ 0, & \bar{r}_W = 0. \end{cases} \quad (6)$$

Smaller T indicates more compact ready-time offsets relative to their mean. When all orders are ready at the release epoch, $T = 0$. The resulting descriptor vector is

$$\Phi(W) = (C(W), I(W), T(W)). \quad (7)$$

3.3.3 Structural classes and within-class selection

The class construction uses the lower and upper tails of C and I ; T records temporal variation separately. For descriptor $X \in \{C, I\}$, let $Q_{u(X)}$ denote its empirical u -quantile over the candidate pool. A common tail probability α defines four candidate-index sets:

$$\begin{aligned} \mathcal{K}_{\text{HC}} &= \{k \in \mathcal{K} : C(W^k) \geq Q_{1-\alpha}(C)\} \\ \mathcal{K}_{\text{LC}} &= \{k \in \mathcal{K} : C(W^k) \leq Q_\alpha(C)\} \\ \mathcal{K}_{\text{HI}} &= \{k \in \mathcal{K} : I(W^k) \geq Q_{1-\alpha}(I)\} \\ \mathcal{K}_{\text{LI}} &= \{k \in \mathcal{K} : I(W^k) \leq Q_\alpha(I)\} \end{aligned} \quad (8)$$

A corner class consists of candidates lying in a tail region of both descriptors. The labels HH, HL, LH, LL record the tails of C and I , respectively, with candidate-index supports

$$\begin{aligned} \mathcal{K}_{\text{HH}} &= \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{HI}} \\ \mathcal{K}_{\text{HL}} &= \mathcal{K}_{\text{HC}} \cap \mathcal{K}_{\text{LI}} \\ \mathcal{K}_{\text{LH}} &= \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{HI}} \\ \mathcal{K}_{\text{LL}} &= \mathcal{K}_{\text{LC}} \cap \mathcal{K}_{\text{LI}} \end{aligned} \quad (9)$$

The selectable labels form $\mathcal{Q} = \{q \in \{\text{HH, HL, LH, LL}\} : \mathcal{K}_q \neq \emptyset\}$. A class-selection instance requires $\mathcal{Q} \neq \emptyset$. Candidates outside all four intersections remain in the candidate pool but outside the corner classes. Because the tail inequalities include their thresholds, quantile ties can produce overlapping supports. Each selectable class is implemented by uniform sampling over its candidate indices. The fixed within-class distribution is

$$\nu_q(k) = \begin{cases} \frac{1}{|\mathcal{K}_q|}, & k \in \mathcal{K}_q, \\ 0, & k \notin \mathcal{K}_q, \end{cases} \quad q \in \mathcal{Q}, k \in \mathcal{K} \quad (10)$$

Selecting class q therefore leads to a draw $\kappa \sim \nu_q$ and the release of W^κ . The class choice determines the distribution of the released wave, while its within-class sampling rule remains fixed.

3.4 Operational evaluation

3.4.1 Processing sequence and AMR assignment

The evaluator reserves resources order by order. For candidate k , let $j \in \mathcal{J}_n = \{1, \dots, n\}$ index positions in the processing sequence

$$\pi(W^k) = (o_1^k, \dots, o_n^k) = \text{sort}_{o \in W^k} (r_o, t_o) \quad (11)$$

Orders are sorted by increasing ready-time offset, with ties resolved by increasing immutable order identifier. All evaluators use this sequence. Each order's complete route is reserved before the next order is considered. Reservation order therefore follows $\pi(W^k)$, while the generated request timestamps need not be globally increasing. Fix a candidate k , evaluator m , and, where applicable, a random realization ξ . These indices are suppressed on operational state quantities below. Let H_a^{j-1}

and G_a^{j-1} be AMR a 's next availability time and associated floor after the first $j-1$ orders have been reserved, initially $H_a^0 = t_w$ and $G_a^0 = f^0$. Order o_j^k is assigned to the earliest-available AMR and begins at

$$a_j = \underset{a \in \mathcal{A}}{\text{lex argmin}}(H_a^{j-1}, a), \quad t_j^0 = \max\{H_{a_j}^{j-1}, t_w + r_{o_j^k}\} \quad (12)$$

The AMR index resolves equal availability times. The maximum in Equation (12) enforces both resource availability and order readiness. An elevator request specifies an origin g , a destination $h \neq g$, and a request time t . Let $\mathbf{S}_m$ contain the elevator reservation state. The request updates this state and returns a completion time:

$$(\hat{t}, \mathbf{S}_m^+) = \mathcal{R}_m(\mathbf{S}_m, g, h, t; \xi) \quad (13)$$

Within a fixed evaluation, $\mathcal{R}_m(g, h, t)$ denotes the returned time $\hat{t}$, with every call replacing $\mathbf{S}_m$ by

$\mathbf{S}_m^+$. The assigned AMR first reaches the order's source and completes pickup:

$$t_j^S = \begin{cases} \mathcal{R}_m(G_{a_j}^{j-1}, s_{o_j^k}, t_j^0), & G_{a_j}^{j-1} \neq s_{o_j^k}, \\ t_j^0, & G_{a_j}^{j-1} = s_{o_j^k}, \end{cases} \quad t_j^P = t_j^S + \tau^P \quad (14)$$

It then reaches the destination and completes delivery:

$$t_j^D = \begin{cases} \mathcal{R}_m(s_{o_j^k}, d_{o_j^k}, t_j^P), & s_{o_j^k} \neq d_{o_j^k}, \\ t_j^P, & s_{o_j^k} = d_{o_j^k}, \end{cases} \quad t_j^C = t_j^D + \tau^D \quad (15)$$

Equations (14) and (15) activate an elevator request only when a movement crosses floors. A same-floor order can still require an initial elevator trip if its assigned AMR is on a different floor. Pickup and drop-off advance the AMR clock by their fixed service durations. After delivery, the assigned AMR becomes available at the destination:

$$H_{a_j}^j = t_j^C, \quad G_{a_j}^j = d_{o_j^k} \quad (16)$$

For every $a \neq a_j$, $H_a^j = H_a^{j-1}$ and $G_a^j = G_a^{j-1}$. Thus, each order updates the state of its assigned

AMR, and subsequent assignments use the resulting availability times.

3.4.2 Throughput abstraction M_1

Evaluator M_1 represents E elevators of capacity c by Ec independent single-request service slots, indexed by $v \in \mathcal{V} = \{1, \dots, Ec\}$. Each slot has a next availability time B_v and a floor G_v associated with that availability, initially $B_v = t_w$ and $G_v = f^0$. A request uses

$$v^* = \underset{v \in \mathcal{V}}{\text{lexargmin}}(B_v, v) \quad (17)$$

Its completion time is

$$\mathcal{R}_{M_1}(g, h, t) = \max\{t, B_{v^*}\} + \gamma |G_{v^*} - g| + \tau^L + \gamma |g - h| + \tau^U \quad (18)$$

Equation (18) combines waiting for availability, empty repositioning to the request origin, loading, travel to the destination, and unloading. The selected slot is updated to $B_{v^*} \leftarrow \mathcal{R}_{M_1}(g, h, t)$ and

$G_{v^*} \leftarrow h$; all other slots retain their states. Capacity is represented by the number of parallel slots, and each request reserves its own trip.

3.4.3 Co-occupancy evaluator M_2

Evaluator M_2 uses E physical cars, each accommodating at most c AMRs on a trip. For each car e , the evaluator stores its next availability B_e , associated floor G_e , and latest reserved trip. That trip is described by its origin-destination pair $(\bar{s}_e, \bar{d}_e)$, loading-end time L_e , completion time D_e , and reserved occupancy P_e . Initially, $B_e = D_e = t_w$, $G_e = f^0$, no trip pair is stored, $L_e = -\infty$, and $P_e = 0$. A request can share a stored trip when its origin and destination match, capacity remains, and it is ready no later than loading ends. The boardable-car set is

$$\mathcal{B}(g, h, t) = \{e \in \mathcal{E} : (\bar{s}_e, \bar{d}_e) = (g, h), P_e < c, t \leq L_e\} \quad (19)$$

If $\mathcal{B}(g, h, t) \neq \emptyset$, the lowest-index eligible car is selected:

$$e^* = \min \mathcal{B}(g, h, t), \quad \mathcal{R}_{M_2}(g, h, t) = D_{e^*}, \quad P_{e^*} \leftarrow P_{e^*} + 1 \quad (20)$$

The joining request shares the trip's stored completion time, and its other reservation quantities remain unchanged. A request that is ready before loading begins can wait for that trip. When $\mathcal{B}(g, h, t) = \emptyset$, the request creates a new trip on the earliest-available car, with equal availability resolved by car index:

$$e^* = \underset{e \in \mathcal{E}}{\text{lex argmin}}(B_e, e) \quad (21)$$

Using the car's pre-update state, the new loading-end and completion times are

$$\begin{aligned} L_{e^*} &= \max\{t, B_{e^*}\} + \gamma |G_{e^*} - g| + \tau^L \\ D_{e^*} &= L_{e^*} + \gamma |g - h| + \tau^U \end{aligned} \quad (22)$$

The elevator returns $\mathcal{R}_{M_2}(g, h, t) = D_{e^*}$ and updates the car's reservation record:

$$B_{e^*} \leftarrow D_{e^*}, \quad G_{e^*} \leftarrow h, \quad (\bar{s}_{e^*}, \bar{d}_{e^*}) \leftarrow (g, h), \quad P_{e^*} \leftarrow 1 \quad (23)$$

Equations (21) to (23) reserve successive trips without overlapping their use of the same car. The new trip replaces that car's stored trip record, and subsequent requests test Equation (19) against the latest record of each car. Here G_e is the floor reached at the next availability time, and P_e counts AMRs reserved on the stored trip.

3.4.4 Stochastic evaluator M_3

Evaluator M_3 retains the car-assignment, boarding, and capacity rules of M_2 . Each new trip receives four independent phase multipliers $Z^{\text{rep}}, Z^{\text{load}}, Z^{\text{trav}}, Z^{\text{unload}}$, corresponding to empty repositioning, loading, loaded travel, and unloading. Multipliers are independent across trips and satisfy

$$\log Z \sim \mathcal{N}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \quad \mathbb{E}[Z] = 1 \quad (24)$$

The loading-end and completion times of a new trip become

$$\begin{aligned} L_{e^*} &= \max\{t, B_{e^*}\} + \gamma |G_{e^*} - g| Z^{\text{rep}} + \tau^L Z^{\text{load}} \\ D_{e^*} &= L_{e^*} + \gamma |g - h| Z^{\text{trav}} + \tau^U Z^{\text{unload}} \end{aligned} \quad (25)$$

The reservation update follows Equation (23). Both a newly reserved request and a joining request return $\mathcal{R}_{M_3}(g, h, t) = D_{e^*}$. Requests joining the same trip share this stored completion time, with no

additional phase draws. Waiting follows from resource availability, while pickup and drop-off durations remain fixed. Let $\xi \in \Xi$ represent the collection of phase multipliers used in a wave evaluation, where Ξ is the realization space. The phase-noise source is independent of the within-class candidate draw κ . Conditional on a candidate and a realization, the reservation relations determine all order completion times.

3.4.5 Wave completion measure

Restoring the candidate, evaluator, and realization indices gives the order completion time $t_{jkm}^C(\xi)$. Wave makespan measures elapsed time from release until the last selected order is delivered:

$$Y_{km}(\xi) = C_{\max}(W^k; m, \xi) = \max_{j \in \mathcal{J}_n} t_{jkm}^C(\xi) - t_W \quad (26)$$

For M_1 and M_2 , the outcome is deterministic and the argument ξ is omitted. Under M_3 , Equation (26) defines the makespan for one phase-time realization.

3.5 Selection formulations

3.5.1 Selection variables

The class formulation chooses a structural class, while the candidate benchmark chooses a particular wave. Table 3.3 lists their variables. Operational assignments and reservation times are supplied by Section 3.4 and determine the performance coefficients used in these formulations.

Table 3.3. Variables in the class and candidate formulations.

Variable	Domain	Definition
z_k	$\{0, 1\}$	1 if candidate wave k is selected
x_o	$\{0, 1\}$	linked indicator that order o belongs to the selected candidate
y_q	$\{0, 1\}$	1 if structural class q is selected
η^C	$\mathbb{R}_+$	auxiliary upper bound on the selected candidate's scores across evaluators
η^Q	$\mathbb{R}_+$	auxiliary upper bound on the selected class's medians across evaluators

AMR assignments, elevator assignments, boarding decisions, and trip times are generated by the fixed evaluator. They are operational state quantities. The tactical optimization variables are z_k , x_o , and y_q .

3.5.2 Robust structural-class selection

The performance of a class is the median makespan induced by its fixed release distribution. When the median is non-unique, we use the midpoint of its median interval. For $q \in \mathcal{Q}$ and $m \in \mathcal{M}$, define

$$\mu_{qm}^{0.5} = \text{Med}_{\kappa \sim \nu_q, \xi} [Y_{km}(\xi)] \quad (27)$$

For M_1 and M_2 , variation in Equation (27) arises from the candidate draw. Under M_3 , the distribution includes both candidate variation and phase-time noise. The median is taken over this joint outcome distribution. Let $\mathcal{M}_D = \{M_1, M_2\}$ be the deterministic evaluator family. The robust class decision minimizes the larger of its two median makespans:

$$q^* \in \arg \min_{q \in \mathcal{Q}} \max_{m \in \mathcal{M}_D} \mu_{qm}^{0.5} \quad (28)$$

An equivalent binary selection formulation is

$$\min_{y, \eta^Q} \eta^Q, \text{ s.t. } \begin{cases} \sum_{q \in \mathcal{Q}} y_q = 1, \\ \eta^Q \geq \sum_{q \in \mathcal{Q}} \mu_{qm}^{0.5} y_q, \\ m \in \mathcal{M}_D, \\ y_q \in \{0, 1\}, \\ q \in \mathcal{Q}, \\ \eta^Q \geq 0 \end{cases} \quad (29)$$

The first constraint selects exactly one nonempty class. The score constraints bound η^Q below by that class's median under each evaluator. Minimization makes η^Q equal to the largest of those medians. The selected class is implemented using Equation (10), and M_3 evaluates its stochastic performance.

3.5.3 Candidate-wave benchmarks

Direct candidate selection uses deterministic scores

$$\ell_{km} = C_{\max}(W^k; m), \quad k \in \mathcal{K}, m \in \mathcal{M}_D \quad (30)$$

For a prespecified nonempty family $\mathcal{M}_R \subseteq \mathcal{M}_D$, the finite-pool benchmark is

$$\min_{k \in \mathcal{K}} \max_{m \in \mathcal{M}_R} \ell_{km} \quad (31)$$

Setting $\mathcal{M}_R = \{m\}$ gives the single-evaluator pool optimum, while $\mathcal{M}_R = \mathcal{M}_D$ gives the two-evaluator robust benchmark. In both cases, the search domain is the given finite candidate pool. The corresponding selection formulation is:

$$\min_{z, x, \eta^C} \eta^C, \text{ s.t. } \begin{cases} \sum_{k \in \mathcal{K}} z_k = 1 \\ x_o = \sum_{k \in \mathcal{K}} \delta_{ok} z_k, o \in \mathcal{O} \\ \sum_{o \in \mathcal{O}} x_o = n \\ \eta^C \geq \sum_{k \in \mathcal{K}} \ell_{km} z_k, m \in \mathcal{M}_R \\ z_k \in \{0, 1\}, k \in \mathcal{K}, x_o \in \{0, 1\}, o \in \mathcal{O}, \eta^C \geq 0 \end{cases} \quad (32)$$

The first constraint selects one candidate. The membership equalities link x_o to that candidate's order set, and the cardinality equality records the fixed wave size. Because every candidate already contains n orders, this cardinality equality is implied by candidate selection and membership. The score constraints and objective then minimize the selected candidate's worst score over $\mathcal{M}_R$.

3.4 Assumptions

A1. Single-wave decision horizon. Each instance begins at t_w and ends when every order in the selected wave is complete. All performance measures refer to this isolated wave, and no other wave overlaps its evaluation interval.

A2. Known candidate information. The source floor, destination floor, ready offset, and immutable identifier of every available order are known when the candidate waves are generated.

A3. Canonical processing order. All evaluators process each candidate wave according to the common sequence $\pi(W^k)$ defined in Equation (1).

A4. Cold-start resource state. At t_w , every AMR and elevator resource is available at floor f^0 .

The instance begins without inherited resource reservations or unfinished orders.

A5. AMR mobility and service. An AMR remains assigned to an order through drop-off and rides the elevator during cross-floor movements. It may therefore serve orders on different floors. The principal evaluator aggregates intra-floor travel and handling into the fixed pickup and drop-off durations τ^P and τ^D . An elevator request is generated whenever the AMR's current floor differs from the next required floor.

A6. Fixed AMR assignment. Orders are considered in canonical order. Each order is assigned to the AMR with the smallest current availability time; an availability tie is resolved by AMR index. Order processing begins at the larger of the assigned AMR's availability time and $t_w + r_o$.

A7. Serial resource reservation. Each order is advanced through its complete route before the next canonical order is considered. Elevator requests update the selected resource state in this reservation order. The generated request timestamps need not be globally increasing.

A8. Stable elevator selection. Equal elevator availability is resolved by virtual-slot index under M_1 and by car index under M_2 and M_3 . When several cars have an open scheduled loading window for the same origin-destination request, the lowest-index boardable car is selected.

A9. Throughput abstraction M_1 . The E elevators with capacity c are represented by Ec independent single-request virtual slots. Each slot has its own location and availability clock. Separate requests occupy separate virtual trips.

A10. Co-occupancy evaluator M_2 . The evaluator contains E physical cars, each with capacity c . A request may join a scheduled trip when its origin and destination match, its request time is no later than the trip's loading-end time, and residual capacity is available. A request that cannot join such a trip initiates a new reservation on the earliest-available car.

A11. Stochastic evaluator M_3 . Evaluator M_3 retains the assignment, boarding, and capacity rules of M_2 . The repositioning, loading, loaded-travel, and unloading durations of every new trip receive mean-one lognormal multipliers that are independent across phases and trips. For each phase multiplier Z ,

$$\log Z \sim \mathcal{N}\left(-\frac{\sigma^2}{2}, \sigma^2\right), \quad \mathbb{E}[Z] = 1. \quad (33)$$

The phase multipliers generated during one complete M_3 wave evaluation form a realization $\xi \in \Xi$, where Ξ is the corresponding realization space. Waiting time follows from the resource availability clock. Pickup and drop-off times retain their fixed values in the principal specification.

A12. Evaluator exogeneity. The resource configuration, operational rule, and evaluator parameters remain fixed while the tactical wave decision changes. Wave composition determines the elevator request stream received by the evaluator.

A13. Fixed within-class implementation. Every selectable class contains at least one candidate. A

selected class is converted to a concrete release by the distribution ν_q in Equation (9). The within-class candidate draw and the M_3 phase-time realization are independent.

3.5 Objective function

3.5.1 Wave makespan

Let $t_{jkm}^C(\xi)$ denote the completion time of the order in canonical position j of candidate k under evaluator m . The realized wave makespan is:

$$Y_{km}(\xi) = C_{\max}(W^k; m, \xi) = \max_{j \in \mathcal{J}_n} t_{jkm}^C(\xi) - t_W \quad (34)$$

The outcome is deterministic under M_1 and M_2 . Under M_3 , ξ indexes the phase-time realization.

3.5.2 Candidate-Wave benchmark

Let $\mathcal{M}_D = \{M_1, M_2\} \subset \mathcal{M}$ denote the deterministic evaluator family. For $m \in \mathcal{M}_D$, define the deterministic candidate score:

$$\ell_{km} = C_{\max}(W^k; m) \quad (35)$$

The single-evaluator finite-pool benchmark is

$$\min_{k \in \mathcal{K}} \ell_{km} \quad (36)$$

For a prespecified nonempty evaluator family $\emptyset \neq \mathcal{M}_R \subseteq \mathcal{M}_D$, the robust candidate benchmark is

$$\min_{k \in \mathcal{K}} \max_{m \in \mathcal{M}_R} \ell_{km} \quad (37)$$

The term pool optimum denotes the optimum obtained over the candidate waves indexed by $\mathcal{K}$.

3.5.3 Robust Class-Selection Objective

For class q and evaluator m , define

$$\mu_{qm}^{0.5} = \text{Med}_{\kappa \sim \nu_q, \xi} [Y_{\kappa m}(\xi)] \quad (38)$$

For M_1 and M_2 , the distribution in Equation (17) is induced by the within-class candidate rule. Under M_3 , it includes both candidate variation and stochastic trip durations. The superscript 0.5 identifies the median estimand. Equation (17) is the primary policy-level estimand. The median-minimax class problem over the deterministic evaluator family is

$$q^* \in \arg \min_{q \in \mathcal{Q}} \max_{m \in \mathcal{M}_D} \mu_{qm}^{0.5} \quad (39)$$

After q^* is selected, the release rule draws $\kappa \sim \nu_{q^*}$. Evaluator M_3 supplies the stochastic performance assessment of the selected class.